CS 8803 LCS: Logic in Computer Science

Fall 2026

Basic Info

Course Number: CS 8803 LCS (CRN 92144)
Class Hours: 12:30pm–1:45pm on Mondays and Wednesdays
Class Room: College of Computing 101

Course Staff:

- Suguman Bansal (website)
  - Instructor
  - suguman@gatech.edu

Course Description

Formal reasoning is rapidly becoming central to frontier AI: in recent weeks alone, OpenAI and Anthropic have both announced new mathematical results whose proofs were formalized and machine-checked in the Lean proof assistant. The logic and automated reasoning that underpin these advances are the subject of this course.

The course provides a rigorous introduction to logic as a foundational tool in computer science, emphasizing formal proof techniques, algorithmic reasoning, and the connections between logic and computational complexity. Topics covered will include:

- Propositional logic
- First-order logic
- Temporal logic
- Computational complexity
- SAT/SMT solving
- Applications in program verification and trustworthy AI

In addition, students will have the opportunity to complete several bonus projects in the Lean proof assistant to formalize theorems taught in class.

Prerequisites

There are no prerequisites for this course. The course is self-contained: all necessary background will be developed in class.

Required Course Materials

No textbook is required. Lecture notes will be provided by the instructor and serve as the primary reference. Supplementary resources will be shared in consultation with students as topics are covered.

Class Format

The class is built around white-board based lectures and is highly interactive. The use of the white-board is intentional: it allows us to be as interactive as possible, so that we learn at the class's speed rather than at the lecture's speed. As a graduate class, we aim to encourage critical thinking, and class discussions are an integral part of the course. Students are encouraged to ask questions, examine assumptions, and engage actively with the material and with one another. The ultimate aim of the course is to build rigorous thinking ability.

Assessments and Grading

This course is graded on a letter grade basis. The final grade will be based on the following components:

[45%] Project 1: SAT Solving Project (Mandatory)
[45%] Project 2: Research project
[10%] Class participation
[Bonus] Bonus Project*: Lean formalizations of course concepts

*The grade earned on the Bonus Project may be used in lieu of the Project 2 grade. It will not contribute towards Project 1 or Class participation.

Course Policies

Academic and Research Honesty/Integrity Statement

Georgia Tech aims to cultivate a community based on trust, academic integrity, and honor. Students are expected to act according to the highest ethical standards. Please review the Student Code of Conduct and the Academic Honor Code, especially Appendix A: Graduate Addendum to the Academic Honor Code.

Any student suspected of cheating or plagiarizing on a quiz, exam, or assignment will be reported to the Office of Student Integrity, who will investigate the incident and identify the appropriate penalty for violations.

Use of AI Tools

Students are welcome to consult AI tools while working on their projects, for example to explore ideas, clarify concepts, or investigate technical questions. Each student remains fully responsible for every submission they turn in, and must be able to clearly explain the content of their work. Submitting material that one cannot explain or defend is inconsistent with the goals of the course and may be treated as a violation of academic integrity.

Collaboration Policy

Students are encouraged to collaborate with one another on projects, and discussing ideas and approaches with classmates is a valuable part of the learning process. However, all submissions must be prepared individually, and each student must be able to explain the work they submit. If you have any questions about what constitutes appropriate collaboration, please ask the instructor.

Accommodations for Students with Disabilities

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services as soon as possible to make an appointment to discuss your special needs and to obtain an accommodations letter. Please also e-mail the instructor as soon as possible in order to set up a time to discuss your learning needs.

Student-Faculty Expectations

At Georgia Tech, we believe that it is important to strive for an atmosphere of mutual respect, acknowledgement, and responsibility between faculty members and the student body. The Student-Faculty Expectations document articulates some basic expectations that you can have of the instructor and that the instructor has of you. In the end, simple respect for knowledge, hard work, and cordial interactions will help build the environment we seek. Students are encouraged to remain committed to the ideals of Georgia Tech while in this class.

Attendance Policy

Regular attendance is expected and contributes to active participation in the course. Since class participation contributes to the final grade, students who must miss a class for genuine reasons are requested to inform the instructor, in advance whenever possible. Students who cannot attend a scheduled lecture are responsible for obtaining notes and catching up on material. Any adjustments to attendance expectations will be communicated in advance.